INCH-POUND

MIL-C-3098/120B(ER)
24 September 1992
SUPERSEDING
MIL-C-3098/120A(EL)
23 June 1975

MILITARY SPECIFICATION SHEET

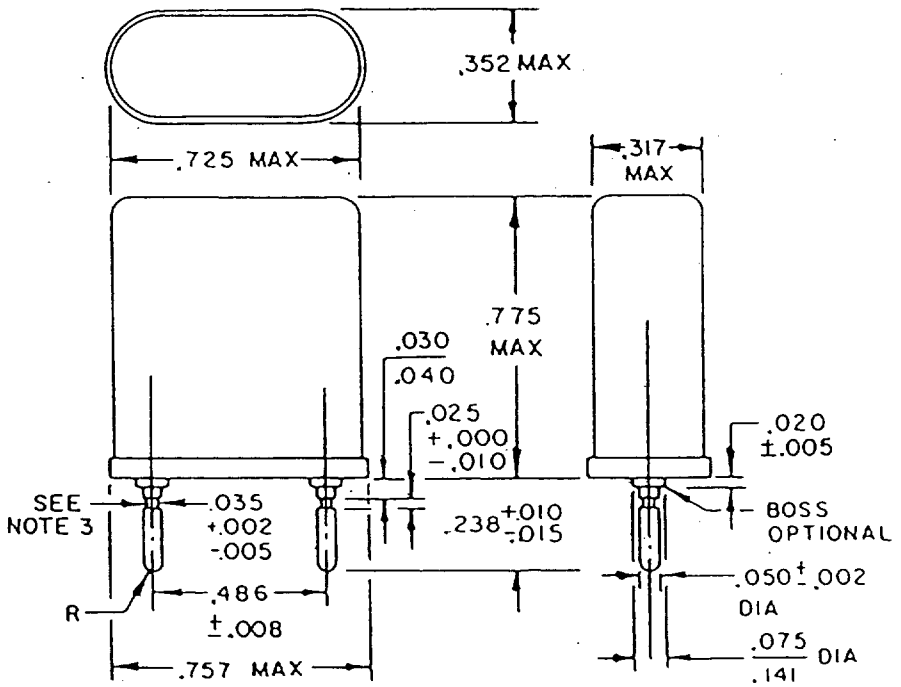
CRYSTAL UNIT, QUARTZ, CR131/U

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; controlled; antiresonance.

INCHES	MM
.002	.05
.005	.13
.008	.20
.010	.25
.015	.38
.020	.51
.025	.64
.030	.76
.035	.89
.040	1.02
.050	1.27
.075	1.91
.141	3.58
.238	6.05
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR131/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-6/U (MIL-H-10056/2).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 to 20 MHz, inclusive

Frequency stability: ±5 ppm.

Antiresonance, load capacitance: 30.0 ±0.5 pF.

Frequency and equivalent resistance:

Reference temperature: 75°C ±1.

Frequency tolerance: ±20 ppm.

Frequency range, inclusive MHz			Maximum resistance ohms
0.8	to	0.85	630
0.85+	to	0.9	600
0.9+	to	1	580
1+	to	1.12	540
1.12+	to	1.25	490
1.25+	to	1.37	450
1.37+	to	1.5	410
1.5+	to	1.62	380
1.62+	to	1.75	330
1.75+	to	1.87	300
1.87+	to	2.0	290
2.0+	to	2.12	270
2.12+	to	2.25	250
2.25+	to	2.6	200
2.6+	to	3	150
3+	to	3.4	110
3.4+	to	3.75	90
3.75+	to	4	75
4+	to	5	60
5+	to	7	35
7+	to	10	24
10+	to	15	22
15+	to	20	20

Mode of oscillation: Fundamental.

Operatable temperature range: -55°C to +70°C and +80°C to +90°C, inclusive.

Operating temperature range: +70°C to +80°C, inclusive.

Drive level: 0.8 to 10.0 MHz = 5.0 ±1.0 mV. 1/

10.0+ to 20.0 MHz = 2.5 ±0.5 mV.

Capacitance, shunt: 7.0 pF maximum.

Shock (specified pulse): 1/

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 to 2.0 MHz ±15 percent.

2.0+ to 20.0 MHz ±10 percent.

1/ The violation of $P = I^2R$ is intentional; at the higher resistance of the crystal unit.

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Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 to 2.0 MHz ±15 percent.

2.0+ to 20.0 MHz ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTES:

1. The socket designed to mate with crystal holder HC-6/U is part no. M12883/04-01 (see MIL-S-12883/4). For Navy use, other sockets listed in MIL-STD-242 may be used.
2. NATO preferred type NXTZ.
3. This crystal unit is identical to the CR27/U, except for a load capacitance rating of 30 ±0.5 pF. This unit is not interchangeable with the CR27/U.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Army - ER

Review activities:
Army - AR, MI
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-A208-01)

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